

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Conditions for life on Earth - Set 1

Mark scheme

AQA spec 3.1.1; The living environment; textbook pp. 5-9

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Mars has low mass so weak gravity and could not retain a dense atmosphere • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen	3
01.2	• Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured • Too close (Venus) - too hot; too far (Mars) - too cold	3
01.3	• The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions)	2
01.4	• The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
02.1	• General physiological solvent: most biological reactions take place between substances dissolved in water • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats	4
02.2	• From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ • The ozone layer reduced UV reaching the surface so organisms could survive out of the water • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out	4
02.3	• Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
03.1	• Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere • Allowing more plant growth, further stabilising climate and soils	3
03.2	<i>Any 3 from:</i> • Lack of ancient historical / instrumental data • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions • Lack of sophisticated equipment / inability to measure many factors • Limited coordination between researchers; lack of data collection in many areas (oceans, poles)	3
03.3	• The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
04.1	• Its mass	1

Question	Marking guidance	Marks
04.2	- The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced - The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions	2
04.3	- The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze - Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
05.1	• $2.7 - 2.45 = 0.25$ billion = 250 million years	1
05.2	- For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans - Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere	2
05.3	<i>Any 3 from:</i> - Oxygen in the atmosphere absorbed ultraviolet light - A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed - Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) - Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out	3
05.4	- Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) - The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years)	2
05.5	<i>Any 2 from:</i> - Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion - Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas - Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
06.1	• Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used	1
06.2	- Age / date of the sample (radioisotope dating) - Past temperatures - Composition of the past atmosphere (eg CO₂ / methane concentrations) - Temperature: more magnesium is incorporated into calcite at higher temperatures	4

Question	Marking guidance	Marks
06.3	• Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present • Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked	2
06.4	<i>Any 3 from:</i> • There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans • Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely • The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen • Water acted as the solvent for reactions / transport	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	